

BrainFace: Towards Identity-Level Face Recognition from Human EEG Signals

Anonymous Author(s)

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Abstract

Face identity recognition is a fundamental cognitive ability of the human brain, yet whether fine-grained identity information can be reliably decoded from non-invasive electroencephalography (EEG) signals remains largely unexplored. Existing EEG visual decoding benchmarks mainly focus on object categories, semantic concepts, or image retrieval, leaving identity-level face decoding without a dedicated benchmark or systematic evaluation protocol. In this work, we introduce BrainFace-40, a multi-subject EEG benchmark designed for identity-level face decoding. BrainFace-40 contains face-evoked EEG recordings across 40 identities and multiple independently recorded sessions, enabling comprehensive evaluation of within-subject decoding, cross-session stability, cross-subject transferability, temporal context effects, and identity-related cognitive factors. To support this task, we further propose EEG-FOFNO, a spectral learning framework that combines Electrode Topology Encoding (ETE) and Weighted Fourier Spectral Encoding (WFSE) with an adaptive fusion module to learn identity-discriminative EEG representations. Extensive experiments demonstrate that facial identity can be decoded from EEG signals substantially above chance level across multiple participants, providing evidence that identity-related neural information is measurable using non-invasive EEG recordings. Further analyses show that these representations exhibit partial temporal stability across independently recorded sessions and partial transferability across subjects, while remaining influenced by temporal context and familiarity-related cognitive factors. In addition, EEG-FOFNO achieves strong performance on EEG-ImageNet and ThingsEEG, suggesting that the proposed topology-spectral representation is not limited to face identity recognition but is also effective for broader EEG-based visual decoding tasks. Overall, BrainFace-40 and EEG-FOFNO provide a new benchmark and baseline for studying fine-grained neural identity representations from EEG signals.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

Electroencephalography, Neural Decoding, EEG Classification, Face Recognition

1 Introduction

Human face recognition is one of the most remarkable cognitive abilities of the brain. Within a few hundred milliseconds after visual exposure, humans can rapidly identify familiar individuals and distinguish among numerous faces despite substantial variations in pose, illumination, expression, and appearance. Extensive studies in cognitive neuroscience have shown that face perception elicits

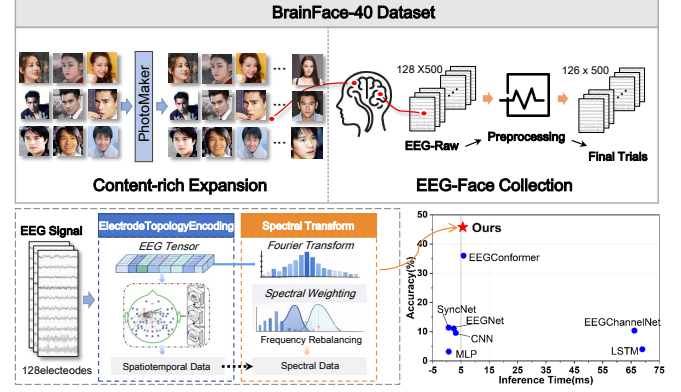


Figure 1: Illustrative overview of the proposed EEG-FOFNO framework and the BrainFace-40 EEG dataset for brain-perceived face recognition.

characteristic neural responses, including the well-known N170 component and distributed activity across occipito-temporal cortical regions [4, 36]. While these findings have greatly advanced our understanding of face perception, a fundamental question remains largely unexplored: can identity-level face information be reliably decoded from non-invasive electroencephalography (EEG) signals?

Recent advances in EEG decoding and deep learning have enabled increasingly accurate classification and reconstruction of visual stimuli from neural recordings. Large-scale benchmarks such as EEG-ImageNet and ThingsEEG have demonstrated that semantic object categories and visual concepts can be decoded from EEG signals, promoting the development of various EEG representation learning methods. However, most existing studies focus on coarse-grained semantic recognition, such as object categories, visual concepts, or image retrieval. In contrast, identity-level face decoding is a substantially more challenging fine-grained problem. Unlike object recognition, where categories often differ in large visual and semantic structures, face identities share highly similar configurations and differ only in subtle appearance details. Therefore, whether EEG signals contain sufficient information to distinguish individual facial identities remains an open question.

Investigating identity-level EEG face decoding is important for both neuroscience and artificial intelligence. From a neuroscience perspective, successful decoding would provide evidence that identity-related neural representations are measurable using non-invasive EEG recordings, offering a new tool for studying human face processing mechanisms. From an artificial intelligence perspective, understanding identity representations in brain signals may contribute to brain-inspired recognition systems and next-generation brain-computer interfaces capable of decoding fine-grained visual information.

Despite its importance, progress in this direction is limited by four major challenges. First, there is no dedicated multi-subject EEG benchmark for identity-level face recognition. Existing EEG datasets mainly target object recognition, image retrieval, motor imagery, or affective computing, leaving the study of face identity representations largely unexplored. Second, EEG recordings exhibit substantial variability across sessions and individuals, making it difficult to determine whether learned representations reflect stable identity-related neural patterns or recording-specific artifacts. Third, temporal context may strongly influence EEG-based face decoding, yet its effect on identity representation remains insufficiently studied. Existing stimulus presentation paradigms often differ substantially in identity continuity, but the impact of such continuity on neural identity decoding is rarely analyzed systematically. Fourth, identity-level decoding requires models to capture subtle neural differences among visually similar faces, which may be difficult for existing EEG architectures that rely mainly on local temporal filtering or generic spatial modeling.

To address these challenges, we introduce BrainFace, a large-scale multi-subject EEG benchmark specifically designed for identity-level face decoding. BrainFace contains EEG recordings collected while participants viewed face images from multiple identities under controlled experimental conditions. Each participant was recorded across multiple independent sessions acquired at different time points, enabling evaluation of both within-subject decoding and temporal generalization. Beyond providing a benchmark dataset, BrainFace is designed to support systematic analysis of identity-related neural representations, including within-subject decoding, cross-session stability, cross-subject transferability, subject scaling behavior, temporal context effects, and cognitive factors such as familiarity.

To facilitate identity-level EEG decoding, we further propose EEG-FOFNO, a parallel spectral learning framework for extracting identity-discriminative EEG representations. EEG-FOFNO consists of two complementary branches, namely Electrode Topology Encoding (ETE) and Weighted Fourier Spectral Encoding (WFSE), followed by an adaptive fusion module for identity prediction. ETE captures topology-aware spatiotemporal features from EEG signals, while WFSE performs full-spectrum frequency-domain modeling with adaptive emphasis on informative spectral components. By integrating these complementary representations, EEG-FOFNO jointly exploits spatial, temporal, and spectral information for fine-grained face identity decoding.

Using BrainFace and EEG-FOFNO, we conduct a systematic investigation of identity-related neural representations in face-evoked EEG signals. Specifically, our study is organized around three fundamental research questions: **(Q1)** Can facial identity be reliably decoded from non-invasive EEG signals? **(Q2)** How do temporal context and session variability influence EEG-based face identity decoding? **(Q3)** To what extent are identity-related neural representations shared across different individuals?

Extensive experiments reveal several findings. First, facial identity can be decoded from EEG signals substantially above chance level across multiple participants, indicating that identity-related information is measurable using non-invasive EEG recordings. Second, temporal context significantly influences decoding performance, suggesting that neural identity representations are shaped

by both stimulus-specific responses and recent perceptual history. Third, although cross-session and cross-subject decoding remain challenging, meaningful identity information persists across independently recorded sessions and unseen participants, suggesting that face identity representations contain both temporally stable and subject-invariant components. Finally, EEG-FOFNO achieves strong performance not only on BrainFace but also on public EEG visual decoding benchmarks, indicating that the proposed topology-frequency representation is useful across multiple EEG-based visual stimulus decoding tasks.

In summary, this work makes the following contributions:

- We introduce **BrainFace**, a large-scale multi-subject EEG benchmark dedicated to identity-level face decoding, enabling systematic study of fine-grained neural face identity representations.
- We propose **EEG-FOFNO**, a topology-frequency EEG representation learning framework that jointly models electrode-topological spatiotemporal dependencies and full-spectrum neural dynamics for identity-level decoding.
- We establish a comprehensive evaluation protocol covering within-subject decoding, cross-session generalization, cross-subject transfer, subject scaling, and identity-wise analysis, providing a unified benchmark for future EEG-based face recognition research.
- We provide comprehensive analyses of identity-related neural representations, showing that facial identity information is decodable, partially stable over time, partially transferable across subjects, and influenced by cognitive factors such as familiarity.

2 Related Work

2.1 EEG Datasets for Visual Decoding

Decoding visually perceived content from Electroencephalography (EEG) signals has become an emerging interdisciplinary field [1, 9, 33]. Compared with fMRI or MEG, EEG offers a number of advantages, including superior temporal resolution, low cost, and portability, making it highly suitable for real-time brain-computer interface (BCI) applications [8, 33]. These advantages make EEG particularly valuable for applications such as cognitive state monitoring, neuro-feedback, and brain-controlled devices. However, EEG's inherently low signal-to-noise ratio (SNR) and strong susceptibility to artifacts (e.g., eye blinks, muscle activity) continue to challenge large-scale data collection and model generalization.

Recent advances in EEG decoding have been driven by datasets with randomized trials such as EEG-ImageNet [35], ThingsEEG [6], and NICE-EEG [33]. These datasets introduced more diverse object stimuli and improved experimental paradigms, enabling neural decoding models to learn more generalized visual representations. However, existing EEG datasets are typically designed around multiple generic object categories (e.g., cats, dogs, and tables), which primarily support category-level recognition. As a result, they fail to adequately capture fine-grained variations within a single semantic category, making them unsuitable for identity-level recognition tasks. In contrast, human faces elicit more consistent and

information-rich EEG responses than generic objects [32], suggesting that face stimuli provide a more suitable foundation for studying identity-related neural representations. Nevertheless, large-scale EEG datasets centered on a single semantic concept remain scarce, limiting further research in this direction.

2.2 EEG Decoding Models

The evolution of EEG decoding models mirrors the broader trajectory of deep learning. Early works relied on classical classifiers such as SVM and LDA [1, 16], which treated EEG channels as independent features and largely ignored temporal and spatial correlations. With the rise of deep learning, Convolutional Neural Networks (CNNs) and their EEG-specific variants, such as EEGNet [18], EEGChannelNet [30], and SyncNet [21], were introduced to capture local spatiotemporal dependencies. Recurrent models LSTM [7] further exploited the sequential nature of EEG signals, while more recent attention-based architectures, such as EEGConformer [34], modeled global dependencies via self-attention. More recently, contrastive cross-modal frameworks represented by Nice-EEG have further extended EEG decoding toward visual representation alignment. Zou [44] combined EEG classification with event-related potential (ERP) analysis to emphasize the interpretive role of neural components in decoding behavior. DAEEGViT [29] further explored Transformer-based EEG modeling by introducing domain adaptation to improve cognitive state identification under inter-domain variability. Despite these advances, existing models still suffer from key limitations. CNNs remain biased toward local receptive fields, potentially missing global coherence across electrodes, whereas Transformers, though effective at modeling long-range dependencies, are computationally expensive and prone to overfitting on EEG datasets, where training samples for each concept are often limited. In addition, most existing models operate mainly in the raw signal domain (time and spatial dimensions), making it difficult to explicitly characterize spectral structures that are central to neural oscillations and frequency-specific brain responses.

2.3 Neural Operators and Fourier Methods

Neural Operators (NOs) [17] have recently emerged as a powerful framework for learning mappings between function spaces, providing resolution-invariant modeling capabilities that generalize across varying input resolutions [42]. Among them, the Fourier Neural Operator (FNO) [23] performs transformations directly in the spectral domain via fast Fourier transforms, enabling efficient modeling of long-range dependencies with significantly reduced complexity in comparison with conventional CNN-based deep models [10, 31].

More recent studies have further extended neural operators beyond the original FNO formulation. Localized operator designs have been introduced to alleviate the tendency of global spectral convolutions to miss fine-scale structures [26], while Fourier-based neural operators on arbitrary domains have improved the applicability of spectral operator learning beyond regular grids [24]. In addition, dynamic Fourier kernels have been proposed to enhance the expressive power of FNOs by allowing input-adaptive spectral interactions [5]. DiffFNO [25] explores the use of Fourier-based operator

learning for arbitrary-scale image super-resolution, demonstrating the ability of spectral operators to capture long-range dependencies in visual signals. Similarly, Arbitrary-scale Fusion Neural Operator [43] applies neural operators to multi-scale visual fusion, highlighting their flexibility in modeling cross-resolution feature interactions. These studies suggest that frequency-domain operator learning can provide complementary benefits to conventional convolutional neural networks, particularly in capturing long-range dependencies and modeling structured signals.

Such properties are especially relevant for EEG data, where both global neural synchronization and fine-grained temporal dynamics need to be represented effectively. However, standard FNOs often truncate high-frequency components to reduce computational cost, which may discard fine-scale details that are particularly important for EEG signals, since both low-frequency global coherence and high-frequency oscillatory activity carry critical neural information. To address this limitation, we propose a learnable frequency rebalancing mechanism, termed the Weighted Fourier Neural Operator (WFNO), to preserve the full spectrum while adaptively weighting different frequency bands. Building upon this design, we further construct a Fusion-Optimized Fourier Neural Operator (EEG-FOFNO) to jointly model local spatiotemporal structure and global spectral dependencies for EEG decoding, achieving state-of-the-art performance on BrainFace-40 and other benchmarks.

3 BrainFace-40 Dataset

3.1 Dataset Overview

BrainFace-40 is a high-resolution EEG dataset specifically designed for brain-perceived face recognition. The dataset contains 100,000 EEG trials collected from five subjects, with 20,000 trials acquired from each subject. Each EEG trial corresponds to the neural response elicited by viewing a unique face image. The visual stimuli consist of 40 celebrity identities (20 male and 20 female), with 500 images per identity spanning substantial appearance variations, including pose, illumination, expression, and background changes. This balanced dataset design mitigates gender bias and provides a robust benchmark for investigating a wide range of EEG-based cognitive decoding tasks, including identity recognition, gender decoding, and neural representation analysis.

EEG recordings were collected from a single healthy adult male participant under controlled laboratory conditions using a 128-channel GREENTEK system. After acquisition, the signals were re-referenced to the linked earlobe electrodes (A1/A2) to reduce common-mode noises, effectively removing the reference pair from the analysis space. As a result, each trial is represented as a 126×500 spatiotemporal matrix, where 126 corresponds to the remaining active electrodes and 500 correspond to the 500 ms post-stimulus window. Unlike prior studies [6] that commonly analyze short (e.g., 100 ms) intervals with limited perceptual relevance, we preserve the entire 0–500 ms post-stimulus duration, following the settings in [35]. This interval robustly captures key face-selective event-related potential (ERP) components, including N170 and P200, which are essential for modeling perceptual and cognitive responses to stimuli of facial identities [4, 36].

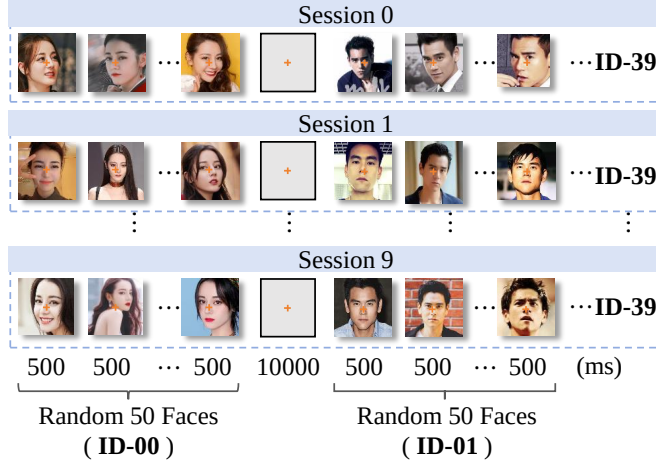


Figure 2: Illustration of the stimulus presentation protocol.

3.2 Stimuli Set and Protocols

To establish a stimuli set for face recognition from EEG trials, we collected 2000 high-quality facial images with 40 identities, each of which has 50 sample images. The image collection is carried out to meet such selection criteria that all faces have clear front views, neutral expression, minimal occlusion, and high recognizability, and some samples are illustrated in Fig. 1 and Fig. 2. To increase the diversity of intra-identities without incurring any negative impact upon the identity recognizability, we adopted a diffusion-model-based content generation pipeline built upon PhotoMaker [22] to expand the 50 sample images of each identity to 500 identity-consistent variants via introduction of subtle but natural variations in facial appearances, including expression, poses, styles, and lighting etc. All synthesized images underwent manual screening to eliminate artifacts or identity drifts. Consequently, the original set (40 identities $\times$ 50 images per identity) was expanded to 40 identities $\times$ 500 images per identity, resulting in a total of 20,000 high-fidelity facial images, which are then presented to stimulate the brains and collect responded EEG signals.

Fig. 2 illustrates an overview of the protocol design for EEG collections, where all facial images are arranged in 10 sessions, within each of which the order of all 40 identities is randomly numbered but kept the same for all sessions. During each session, however, all 500 images are randomly ordered to ensure that the collected EEG signals represent random trials the same as that reported by Ahmed et al [1]. As seen in Fig. 2, a central bull's eye fixation target (“+”) was present on the screen throughout the entire experiment [38], and we asked participants to constantly gaze at it. In each trial, a target face was displayed at the center of the screen for 500 ms, followed by the sequential presentation of 50 different images of the same identity. After completing one identity block (50 images), participants took a $10 \times 1000\text{ms}$ rest to prevent visual fatigue. The next block then introduced a new identity with a distinct set of randomized facial images.

A total of 40 identities were presented per session, forming one complete recording cycle. The experiment included 10 sessions,

with non-overlapping image sets across sessions to ensure stimulus independence. Importantly, the 10 sessions were collected at independent time points rather than being segmented from a single continuous recording, thereby reflecting genuine temporal variability across recording runs, while maximizing sustained attention [3, 27].

3.3 EEG Recording and Preprocessing

EEG signals were acquired using a 128-channel GREENTEK system arranged under the 10–10 international standard [28], amplified via BrainVision actiCHamp Plus. The raw datasets were sampled at 5,000 Hz, then downsampled to 1,000 Hz for analysis. The preprocessing involved include: (i) rereferencing to A1/A2 (earlobe electrodes); (ii) zero-phase Butterworth filtering (0.3–40 Hz) with a 50 Hz notch filter; (iii) ocular artifact removal using Independent Component Analysis (ICA); (iv) semi-automatic rejection of residual noises; (v) epoch segmentation from 0 ms to +500 ms relative to stimulus onset; (vi) baseline correction over the pre-stimulus interval; and (vii) multivariate noise normalization.

The resulting dataset contains 100,000 randomized EEG trials acquired from five subjects (20,000 trials per subject). Each trial is represented as a spatiotemporal tensor of size (126×500) , corresponding to 126 EEG channels and 500 temporal sampling points.

4 The Proposed EEG-FOFNO

4.1 Overview

We propose **EEG-FOFNO**, a parallel topology–frequency neural operator framework for identity-level face decoding from EEG signals. Given an EEG trial $\mathbf{X} \in \mathbb{R}^{C \times T}$, where C and T denote the number of electrodes and temporal samples, respectively, the goal is to predict the identity label y from the face-evoked neural response. This task is challenging because identity-discriminative EEG patterns are typically weak, noisy, and distributed across both temporal and spectral dimensions. In particular, the neural evidence for face identity may not appear at a single fixed latency, but can be spread over multiple processing stages, including early visual responses, intermediate facial-structure processing, and later identity-related neural activity. Therefore, an effective EEG decoding model should jointly capture local spatiotemporal responses, long-range temporal dependencies, and global frequency-domain dynamics.

As shown in Fig. 3, EEG-FOFNO consists of four major components: an EEG patch embedding module, a Transformer encoder branch, a Fourier neural operator branch, and a lightweight classification head. First, the raw EEG signal is projected into a sequence of compact neural tokens by a convolutional patch embedding module. Specifically, temporal convolution is used to extract short-range waveform patterns, while spatial convolution across electrodes aggregates multi-channel neural responses into topology-aware EEG representations. A temporal average-pooling operation then partitions the continuous EEG response into a sequence of local temporal patches. This produces an embedded EEG sequence

$$\mathbf{Z}_0 = \phi_{\text{emb}}(\mathbf{X}) \in \mathbb{R}^{N \times d}, \quad (1)$$

where N denotes the number of EEG patches and d is the hidden feature dimension. This design transforms the high-dimensional

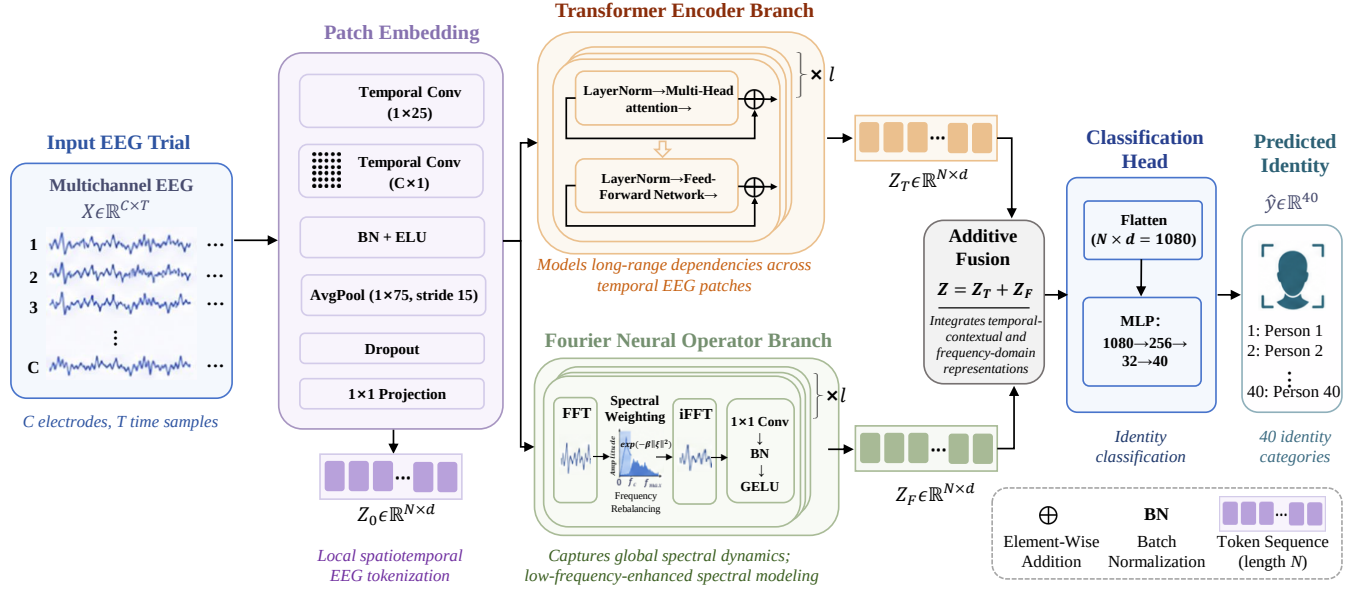


Figure 3: Structural illustration of the proposed EEG-FOFNO, which consists of two core components: Electrode Topology Encoding and Weighted Fourier Spectral Encoding, along with an Adaptive Fusion Module for brain-perceived face recognition.

and noisy raw EEG recording into a more compact token sequence, making it suitable for subsequent sequence modeling.

To model distributed identity-related neural responses, EEG-FOFNO processes Z_0 with two complementary branches. The first branch is a Transformer encoder, which models the contextual dependency among EEG patches:

$$Z_T = \phi_T(Z_0). \quad (2)$$

Through multi-head self-attention, each EEG patch can adaptively attend to other temporal patches. This is important for face-evoked EEG decoding because identity information may be jointly encoded by multiple temporal stages rather than by an isolated response peak. In this sense, the Transformer branch learns how neural evidence from different time segments should be integrated for identity discrimination.

In parallel, the second branch introduces a Fourier neural operator to capture global spectral structures:

$$Z_F = \phi_F(Z_0). \quad (3)$$

Unlike purely time-domain operations, the Fourier branch maps EEG representations into the frequency domain, applies learnable spectral transformations, and then projects them back to the representation space. This enables the model to explicitly characterize global frequency-domain dependencies in face-evoked EEG signals. Since EEG responses naturally contain rhythmic and frequency-specific neural dynamics, spectral modeling provides complementary information to the temporal attention mechanism. In particular, the Fourier branch is designed to emphasize stable low-frequency neural components while suppressing less reliable high-frequency fluctuations, which is beneficial for robust EEG-based identity decoding.

The outputs of the two branches are then integrated by residual additive fusion:

$$Z = Z_T + Z_F. \quad (4)$$

This simple fusion strategy preserves the original token structure and allows the model to combine temporal-contextual information from the Transformer branch with global spectral information from the Fourier branch without introducing excessive parameters. The fused representation Z is finally flattened and passed into a multi-layer classification head:

$$\hat{y} = \phi_{\text{cls}}(\text{Flatten}(Z)), \quad (5)$$

where $\hat{y} \in \mathbb{R}^K$ denotes the predicted logits over K identity categories.

Overall, EEG-FOFNO is designed around the intrinsic properties of face-evoked EEG signals. The patch embedding module extracts local spatiotemporal neural responses from raw multi-electrode recordings; the Transformer encoder captures long-range dependencies among temporally distributed EEG patches; the Fourier neural operator models global spectral dynamics that are difficult to learn using purely local convolution or attention alone; and the final fusion mechanism integrates these complementary representations for fine-grained identity classification. By jointly exploiting temporal context and frequency-domain structure, EEG-FOFNO provides a compact yet expressive architecture for decoding individual face identities from non-invasive EEG signals.

4.2 EEG Patch Embedding

EEG signals contain rich temporal neural dynamics and cross-channel dependencies induced by distributed brain activity. To capture these properties while reducing the redundancy and noise of raw recordings, EEG-FOFNO first employs an EEG patch embedding module to transform each trial into a compact sequence of

Table 1: Network configuration of the Electrode Topology Encoding (ch : electrodes, k : hidden channels, m : temporal patches).

Layer	Input	Output	Kernel	Stride
Temporal Aggregation	1	k	(1, 25)	(1, 1)
Electrode Aggregation	k	k	(ch , 1)	(1, 1)
Avg Pooling	k	k	(1, 75)	(1, 15)
Rearrange	$(k, 1, m)$	$\rightarrow (m, k)$	–	–

neural tokens. Given an EEG trial $\mathbf{X} \in \mathbb{R}^{C \times T}$ with C electrodes and T temporal samples, the embedding module maps the raw signal into

$$\mathbf{Z}_0 = \phi_{\text{emb}}(\mathbf{X}) \in \mathbb{R}^{N \times d}, \quad (6)$$

where N denotes the number of temporal EEG patches and d is the embedding dimension.

As summarized in Table 1, the patch embedding module consists of temporal filtering, electrode-wise spatial aggregation, nonlinear normalization, temporal pooling, and point-wise feature projection. We first apply a temporal convolution with kernel size (1, 25) along the time dimension. This operation extracts local waveform patterns from short EEG segments and captures short-term neural responses that may reflect task-relevant temporal dynamics elicited by face stimuli. By operating directly on the temporal axis, the model preserves the sequential structure of EEG signals while learning discriminative local response features. We then perform electrode-wise spatial aggregation using a convolution with kernel size (C , 1) across all EEG channels. Rather than explicitly imposing a predefined electrode geometry, this operation provides a data-driven global channel mixing mechanism, enabling the network to learn coordinated neural responses distributed across multiple electrodes. This is particularly important for face-evoked EEG decoding, where identity-related neural evidence is often weak and spatially distributed rather than confined to a single channel. Due to the high temporal resolution of EEG recordings, the convolutional features are densely sampled along the time dimension and may contain redundant local activations. To improve representation compactness and enlarge the effective temporal context, we employ an average pooling layer with kernel size (1, 75) and stride (1, 15) to aggregate neighboring temporal features. This operation also serves as an implicit patch slicing mechanism, converting the continuous EEG response into a sequence of local temporal tokens. Dropout is further introduced to reduce overfitting to trial-specific fluctuations and improve robustness to EEG noise. Finally, a 1×1 projection maps the pooled features into a fixed-dimensional embedding space, followed by a rearrangement operation that produces the token sequence $\mathbf{Z}_0$. The resulting representation retains local temporal dynamics while incorporating cross-channel neural dependencies, providing a compact and noise-robust input for the subsequent temporal and spectral representation learning branches.

4.3 Transformer Encoder Branch

After patch embedding, each EEG trial is represented as a sequence of temporal neural tokens $\mathbf{Z}_0 \in \mathbb{R}^{N \times d}$, where each token encodes

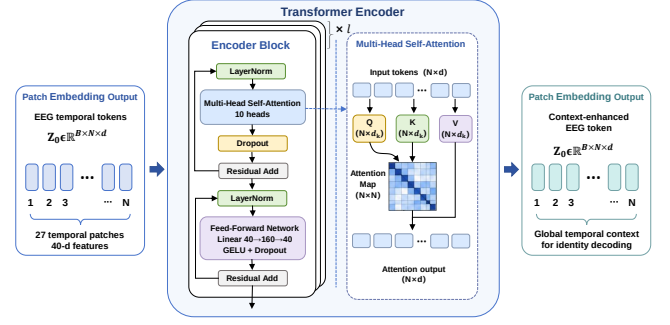


Figure 4: Illustration of the Transformer encoder branch. The self-attention mechanism enables each temporal EEG patch to attend to all other patches, thereby capturing long-range temporal dependencies for identity-level EEG decoding.

local spatiotemporal responses within a short EEG segment. However, identity-related neural evidence in face-evoked EEG is rarely confined to a single time window. Instead, it may be distributed across multiple processing stages, including early visual responses, intermediate face-structure processing, and later identity-sensitive neural activity. To model such temporally distributed dependencies, EEG-FOFNO employs a Transformer encoder branch to capture contextual interactions among EEG patches.

As illustrated in Fig. 4, the Transformer branch consists of L stacked encoder blocks. Each block follows a pre-normalized residual design, including a multi-head self-attention module and a position-wise feed-forward network. For the ℓ -th encoder block, the token sequence is updated as

$$\tilde{\mathbf{Z}}_\ell = \mathbf{Z}_{\ell-1} + \text{MSA}(\text{LN}(\mathbf{Z}_{\ell-1})), \quad (7)$$

$$\mathbf{Z}_\ell = \tilde{\mathbf{Z}}_\ell + \text{FFN}(\text{LN}(\tilde{\mathbf{Z}}_\ell)), \quad (8)$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\text{MSA}(\cdot)$ denotes multi-head self-attention, and $\text{FFN}(\cdot)$ denotes the feed-forward network. The residual connections preserve the original token information and stabilize optimization when learning from noisy EEG representations.

The self-attention mechanism allows each EEG token to adaptively attend to all other temporal tokens. This global token interaction is important for face-evoked EEG decoding because identity-discriminative responses may vary in latency across trials and subjects. Instead of relying on a fixed local receptive field, the Transformer branch dynamically aggregates neural evidence from different temporal segments and suppresses less informative or noisy activations. The feed-forward network further refines each token in the feature space, enhancing non-linear discrimination after temporal context aggregation.

After L encoder blocks, the Transformer branch produces the context-enhanced EEG representation

$$\mathbf{Z}_T = \phi_T(\mathbf{Z}_0) \in \mathbb{R}^{N \times d}. \quad (9)$$

This representation encodes long-range temporal dependencies among EEG patches and serves as the temporal-contextual branch for subsequent fusion with the Fourier neural operator representation.

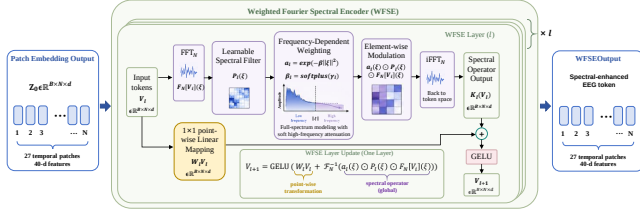


Figure 5: Illustration of the proposed Weighted Fourier Spectral Encoder (WFSE) branch.

4.4 Weighted Fourier Spectral Encoder Branch

The Transformer encoder branch models contextual interactions among temporal EEG patches in the token space. However, EEG signals also contain rich frequency-specific neural dynamics, and identity-related responses may be reflected not only by temporal dependencies but also by global spectral patterns. To explicitly capture such frequency-domain information, we propose a **Weighted Fourier Spectral Encoder (WFSE)** branch. As illustrated in Fig. 5, WFSE is not a direct application of a conventional Fourier Neural Operator (FNO) [23]; instead, it is designed for EEG-based identity decoding by combining full-spectrum spectral transformation with a learnable frequency-dependent weighting mechanism.

Let $V_0 = Z_0 \in \mathbb{R}^{N \times d}$ denote the embedded EEG token sequence, where N is the number of temporal patches and d is the hidden dimension. For the l -th WFSE layer, we first transform the input tokens into the Fourier domain along the temporal-patch dimension:

$$\widehat{V}_l = \mathcal{F}_N[V_l], \quad (10)$$

where $\mathcal{F}_N(\cdot)$ denotes the one-dimensional Fourier transform over the token sequence. In the spectral domain, a learnable complex-valued spectral filter $P_l(\xi)$ is used to model global frequency interactions, where ξ denotes the frequency index.

Conventional FNO implementations often retain only a fixed number of low-frequency modes for computational efficiency. While this hard truncation is effective in many physical systems, it may be suboptimal for EEG-based identity decoding. Discarding all higher-frequency components may remove fine-grained neural cues, whereas retaining the full spectrum without regularization can introduce noise-sensitive spectral responses. To balance these factors, WFSE retains the full spectral representation and introduces a learnable frequency-dependent weighting function:

$$a_l(\xi) = \exp(-\beta_l \|\xi\|_2^2), \quad (11)$$

where $\beta_l > 0$ controls the degree of spectral decay. We parameterize it as

$$\beta_l = \text{softplus}(\gamma_l), \quad (12)$$

with γ_l being an unconstrained learnable scalar. This parameterization guarantees positive spectral decay while allowing each WFSE layer to adaptively determine how strongly high-frequency components should be attenuated.

The weighted spectral operator is then formulated as

$$\mathcal{K}_l(V_l) = \mathcal{F}_N^{-1} \left(a_l(\xi) \odot P_l(\xi) \odot \widehat{V}_l(\xi) \right), \quad (13)$$

where $\mathcal{F}_N^{-1}(\cdot)$ denotes the inverse Fourier transform and $\odot$ represents element-wise spectral modulation. The complete WFSE

layer combines this global spectral operator with a local point-wise transformation:

$$V_{l+1} = \sigma(\mathcal{W}_l V_l + \mathcal{K}_l(V_l)), \quad (14)$$

where $\mathcal{W}_l$ is a point-wise linear mapping and $\sigma(\cdot)$ denotes a non-linear activation function. The spectral branch captures global temporal-frequency dependencies, while the point-wise branch preserves local feature transformations and stabilizes optimization.

This design makes WFSE particularly suitable for EEG decoding. Low-frequency components in EEG are often more stable across trials, whereas high-frequency responses may contain both useful fine-grained information and noise-sensitive fluctuations. Instead of applying a fixed handcrafted filter or hard frequency truncation, WFSE softly regularizes the spectral representation through a learnable decay function. Therefore, the model can retain potentially discriminative high-frequency cues while suppressing unreliable spectral variations.

By stacking multiple WFSE layers, the spectral branch progressively refines frequency-domain EEG representations at different levels of abstraction. The final output of the WFSE branch is denoted as

$$Z_F = \phi_F(Z_0) \in \mathbb{R}^{N \times d}. \quad (15)$$

The resulting representation encodes global spectral dynamics of face-evoked EEG signals and provides complementary frequency-domain information to the temporal-contextual representation learned by the Transformer encoder branch.

4.5 Identity Prediction and Optimization

After the Transformer encoder branch and the WFSE branch, EEG-FOFNO obtains two complementary representations: the temporal-contextual representation $Z_T \in \mathbb{R}^{N \times d}$ and the spectral-enhanced representation $Z_F \in \mathbb{R}^{N \times d}$. Since the two branches operate on the same embedded EEG token sequence and preserve the same token structure, we integrate them through simple additive aggregation:

$$Z = Z_T + Z_F. \quad (16)$$

The fused representation Z is then flattened and fed into a light-weight multilayer classification head to predict the identity category:

$$\hat{y} = \phi_{\text{cls}}(\text{Flatten}(Z)), \quad (17)$$

where $\hat{y} \in \mathbb{R}^K$ denotes the predicted logits over K identity classes. In our task, $K = 40$.

We train EEG-FOFNO in an end-to-end manner using the standard cross-entropy loss. Given the ground-truth identity label y and the predicted class probability

$$p_k = \frac{\exp(\hat{y}_k)}{\sum_{j=1}^K \exp(\hat{y}_j)}, \quad (18)$$

the optimization objective is defined as

$$\mathcal{L}_{\text{CE}} = -\log p_y. \quad (19)$$

This objective directly encourages the fused temporal-spectral EEG representation to be discriminative for identity-level face decoding.

Table 2: Comparative listing of Top-1 and Top-5 accuracies achieved on BrainFace-40 (mean $\pm$ std).

Method(%)	Fold1		Fold2		Fold3		Fold4		Fold5		AVG	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
CNN[20]	9.0 $\pm$ 0.6	34.3 $\pm$ 0.9	9.8 $\pm$ 0.4	35.2 $\pm$ 1.3	9.2 $\pm$ 0.7	33.8 $\pm$ 0.8	9.5 $\pm$ 0.5	33.3 $\pm$ 1.7	9.5 $\pm$ 0.8	33.8 $\pm$ 1.2	9.6	34.1
LSTM[7]	3.9 $\pm$ 0.3	16.5 $\pm$ 0.4	4.3 $\pm$ 0.3	17.4 $\pm$ 1.3	3.7 $\pm$ 0.6	15.8 $\pm$ 1.6	3.9 $\pm$ 0.1	16.0 $\pm$ 0.9	4.2 $\pm$ 0.4	16.6 $\pm$ 1.3	4.0	16.5
MLP[37]	3.1 $\pm$ 0.3	13.6 $\pm$ 0.7	3.2 $\pm$ 0.2	13.5 $\pm$ 1.1	3.4 $\pm$ 0.3	13.3 $\pm$ 1.2	3.1 $\pm$ 0.2	13.3 $\pm$ 0.6	3.2 $\pm$ 0.4	13.8 $\pm$ 1.4	3.2	13.5
EEGNet[18]	11.7 $\pm$ 0.5	47.8 $\pm$ 1.1	10.8 $\pm$ 0.5	47.9 $\pm$ 0.8	11.5 $\pm$ 0.7	47.5 $\pm$ 0.8	10.7 $\pm$ 0.5	47.7 $\pm$ 0.9	10.6 $\pm$ 0.4	45.6 $\pm$ 1.3	11.1	47.3
EEGChannelNet[30]	9.9 $\pm$ 0.4	36.8 $\pm$ 0.8	10.1 $\pm$ 0.5	37.7 $\pm$ 0.9	9.4 $\pm$ 0.4	35.6 $\pm$ 0.8	9.2 $\pm$ 0.4	34.0 $\pm$ 0.7	13.5 $\pm$ 0.6	45.6 $\pm$ 0.9	10.4	37.9
SyncNet[21]	11.4 $\pm$ 0.5	45.2 $\pm$ 0.9	11.2 $\pm$ 0.7	44.0 $\pm$ 1.0	11.4 $\pm$ 0.5	43.7 $\pm$ 0.8	11.5 $\pm$ 0.3	45.0 $\pm$ 0.7	11.4 $\pm$ 0.6	44.3 $\pm$ 1.3	11.4	44.4
EEGConformer[34]	34.0 $\pm$ 0.9	89.7 $\pm$ 1.2	36.4 $\pm$ 1.1	90.9 $\pm$ 1.4	35.8 $\pm$ 1.0	92.3 $\pm$ 1.6	37.1 $\pm$ 0.6	91.5 $\pm$ 1.5	36.7 $\pm$ 1.3	91.7 $\pm$ 1.5	36.0	91.2
DAEEGViT[29]	36.3 $\pm$ 1.1	91.4 $\pm$ 1.7	35.4 $\pm$ 1.4	92.1 $\pm$ 1.8	34.3 $\pm$ 1.4	94.4 $\pm$ 1.8	38.1 $\pm$ 1.3	91.1 $\pm$ 1.5	36.2 $\pm$ 1.8	92.1 $\pm$ 1.6	36.6	92.2
EEG-FOFNO (Ours)	46.6 $\pm$ 1.2	95.0 $\pm$ 1.6	46.1 $\pm$ 1.1	94.5 $\pm$ 1.3	45.8 $\pm$ 1.1	94.7 $\pm$ 1.3	45.3 $\pm$ 1.0	94.0 $\pm$ 1.4	45.2 $\pm$ 1.4	94.3 $\pm$ 1.7	45.8	94.5

5 Experiments

5.1 Experimental Roadmap

The experiments are organized to systematically examine the existence, contextual sensitivity, temporal stability, cross-subject generalizability, and cognitive characteristics of identity-related neural representations in face-evoked EEG signals.

We first conduct within-subject identity decoding experiments (Section 5.2) to address the fundamental question of whether facial identity information can be reliably decoded from EEG recordings within individual participants. This evaluation serves as the primary validation of the proposed EEG-based identity recognition task by determining whether identity-discriminative neural patterns are present in face-evoked EEG responses.

We then investigate temporal context effects (Section 5.3) by varying the Identity Continuity Length (ICL) during stimulus presentation. This analysis examines whether decoding performance is driven solely by stimulus-evoked responses or is also influenced by the temporal continuity of identity presentation, thereby providing insight into the role of temporal structure in shaping identity-related neural representations.

Next, we evaluate cross-session generalization (Section 5.4) to assess the temporal stability of the learned representations. Under the leave-one-session-out protocol, models are trained on multiple recording sessions and tested on an independently acquired temporally unseen session. This setting evaluates whether identity-related neural patterns remain stable across different recording time points.

We further examine cross-subject generalization (Section 5.5) to determine whether identity-related EEG representations contain components that can transfer across individuals. Specifically, we adopt a leave-one-subject-out protocol, where models are trained on four participants and directly evaluated on the remaining unseen participant without subject-specific adaptation.

Finally, we perform cognitive factors analysis (Section 5.6) to investigate potential factors that may modulate identity-specific neural responses, including the gender of the target face and the familiarity between participants and facial identities. Together, these experiments provide a comprehensive evaluation of BrainFace-40 and EEG-FOFNO from the perspectives of task validity, temporal context, temporal robustness, subject transferability, and cognitive interpretability.

5.2 Within-Subject Identity Decoding

We first investigate the most fundamental question of this work: whether face-evoked EEG signals contain sufficient information to support identity-level recognition within individual participants.

To obtain a comprehensive and statistically reliable evaluation, we adopt a subject-wise 5-fold cross-validation protocol for each participant. Specifically, each fold uses 80% of the samples for training and the remaining 20% for testing. The splits are constructed to maintain balanced distributions of face identities, genders, and visual appearance variations across folds, thereby ensuring stable and reproducible evaluation.

(i)Overall Recognition Performance. We begin by analyzing the results of a representative participant. Table 2 summarizes the recognition performance of EEG-FOFNO and all competing methods. Notably, all evaluated methods achieve performance substantially above the chance level, indicating that discriminative identity information is reliably encoded in EEG signals and that the proposed decoding task is both meaningful and learnable. Traditional architectures, including CNN [20], LSTM [7], and MLP [37], achieve relatively modest performance, with the best-performing model reaching only 9.6% Top-1 accuracy and 47.3% Top-5 accuracy. In contrast, EEG-specific architectures [18, 29, 34] provide a substantial performance boost, achieving up to 38.6% Top-1 accuracy and 92.2% Top-5 accuracy. EEG-FOFNO further establishes a clear advantage over all competing methods, surpassing the strongest baseline by 9.8 percentage points in Top-1 accuracy. These results demonstrate that jointly modeling topology-aware spatiotemporal interactions and global spectral dependencies leads to more discriminative EEG representations for identity decoding.

To further investigate the structure of the learned neural representations, Fig. 6 visualizes the test EEG features using t-SNE. Several important observations can be made. First, samples belonging to the same identity generally form compact clusters, while different identities occupy distinct regions in the embedding space. This result provides evidence that face-evoked EEG signals contain discriminative identity-related information and that EEG-FOFNO successfully transforms these signals into identity-separable neural representations. Second, many identities exhibit continuous manifold-like structures rather than isolated point clusters, suggesting that the learned representation preserves intra-identity variations while maintaining identity consistency.

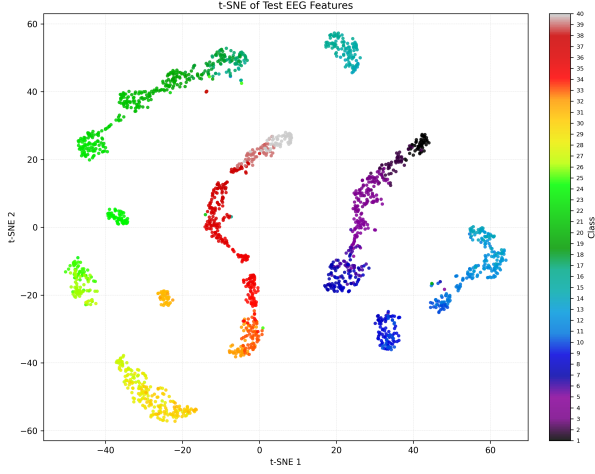


Figure 6: t-SNE visualization of the learned EEG representations for 40 face identities.

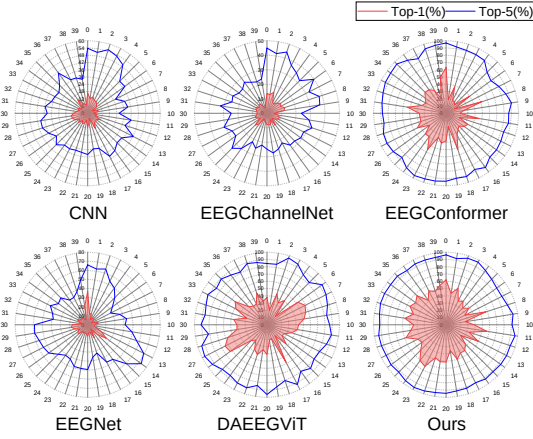


Figure 7: Comparative radial-plot visualization of face recognition performances across 40 facial identities.

Nevertheless, not all identities are perfectly separated. Several neighboring clusters remain partially overlapping or exhibit relatively small inter-class margins. Such overlap is expected given the low signal-to-noise ratio of EEG recordings and the high visual similarity among certain facial identities. Importantly, these ambiguities are localized rather than global, with the majority of identities remaining well separated in the feature space.

(ii) Performance Across Different Identities. While the overall recognition accuracy reflects the average decoding capability of each model, it does not reveal whether the model performs consistently across different identities. We therefore further analyze the identity-wise Top-1 and Top-5 accuracies over all 40 facial identities. As shown in Fig. 7, conventional CNN-based models, including CNN, EEGChannelNet, and EEGNet, exhibit highly irregular radial profiles, indicating substantial performance variation across identities. EEG-based methods improve the overall distribution to some

extent. EEGConformer and DAEeGVIT produce more competitive Top-5 profiles, but their Top-1 curves still show noticeable fluctuations across identities, suggesting that the learned representations remain sensitive to identity-specific variability. In contrast, EEG-FOFNO presents a larger and more balanced Top-1 region, together with a consistently strong Top-5 distribution. The smoother and more expanded radial profile indicates that EEG-FOFNO achieves not only higher average decoding performance but also more uniform identity-wise generalization.

These results further reveal an important property of identity-level EEG decoding. Although Top-1 accuracy still varies across different identities, indicating that some facial identities are intrinsically more difficult to decode from EEG responses than others, the Top-5 profiles of EEG-based methods are generally more stable across identities. This suggests that, even when the model fails to make the correct identity as its top prediction, the discriminative neural evidence is often sufficient to place the correct identity among the top-ranked candidates.

(iii) Performance Across Different Participants. While the above results establish the feasibility of identity decoding for an individual participant, an important question remains: is this phenomenon reproducible across different individuals? To answer this question, we repeat the same evaluation protocol for all five participants where training and testing are performed using data from the same subject under a 5-fold cross-validation setting. The results are summarized in Table 3. Across all participants, EEG-FOFNO consistently achieves substantially above-chance performance and remains the best-performing method. Although performance varies across subjects due to differences in neural responses and recording conditions, the overall trend is remarkably stable, with Top-1 accuracies ranging from 35.6% to 45.8%.

These findings suggest that the existence of decodable identity-related information is not unique to a particular participant but can be consistently observed across multiple individuals. This provides strong evidence that face identity is reliably encoded in EEG signals under within-subject settings.

Table 3: Within-subject identity decoding performance across five participants under the 5-fold cross-validation protocol.

Method	Subj 1	Subj 2	Subj 3	Subj 4	Subj 5
CNN [20]	9.6 (34.1)	9.9 (35.9)	8.4 (30.3)	9.3 (32.7)	8.1 (28.8)
EEGNet [18]	11.1 (47.3)	10.4 (45.3)	8.8 (35.8)	9.9 (44.2)	10.1 (45.1)
EEGChannelNet [30]	10.4 (37.9)	8.2 (31.5)	9.1 (39.7)	8.9 (37.4)	9.3 (42.1)
SyncNet [21]	11.4 (44.4)	11.1 (41.9)	12.3 (55.1)	11.5 (48.7)	10.4 (41.1)
EEGConformer [34]	36.0 (91.2)	31.1 (78.3)	29.6 (81.8)	33.2 (89.1)	29.3 (78.3)
EEG-FOFNO (Ours)	45.8 (94.5)	39.9 (86.5)	35.6 (75.4)	41.4 (88.9)	37.5 (85.8)

5.3 Temporal Context Effects

The previous experiment establishes that facial identity can be reliably decoded from EEG signals. An important question, however, is whether such decoding performance is determined solely by stimulus-evoked neural responses or is also influenced by the temporal continuity of identity presentation.

To investigate this issue, we conduct a controlled study by systematically varying the Identity Continuity Length (ICL) while

keeping all other experimental factors unchanged. Specifically, ICL- N denotes that N consecutive images belonging to the same identity are presented before switching to another identity. We consider five settings: ICL-1, ICL-50, ICL-100, ICL-250, and ICL-500. Notably, ICL-1 and ICL-500 correspond to two opposite extremes, representing complete identity randomization and maximal identity continuity, respectively. Figure 8 summarizes the decoding performance under different continuity levels. A clear trend emerges: recognition accuracy gradually improves as the continuity length increases. This observation suggests that temporal context contributes to identity decoding, likely by facilitating the formation of more stable identity-related neural representations across consecutive presentations of the same individual.

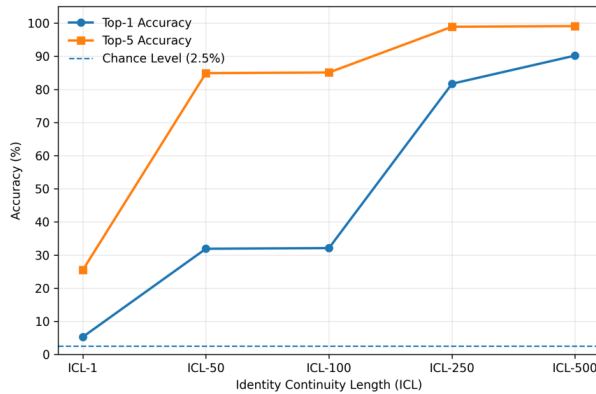


Figure 8: Effects of identity continuity length (ICL) on EEG-based face identity decoding performance.

Importantly, however, identity-related information remains decodable even under the most challenging ICL-1 condition, where the identity changes after every image and temporal continuity is minimized. The performance remains substantially above the 2.5% chance level, indicating that successful decoding cannot be explained solely by contextual effects and that face-evoked EEG signals contain intrinsic identity-related information. At first glance, this may suggest that extremely long continuity conditions such as ICL-500 are preferable. However, our goal is not to maximize contextual facilitation, but rather to balance identity representation stability and contextual bias. ICL-1 and ICL-500 have both been employed in previous EEG-based visual decoding studies, representing two opposite extremes of identity continuity [1, 35]. ICL-1, minimizes temporal dependency but introduces frequent identity switching, making it difficult to establish stable identity-related neural representations. In contrast, ICL-500 provides the strongest continuity but may increase the influence of long-range contextual effects and block-related confounds.

Taken together, these findings suggest that neither fully randomized presentation (ICL-1) nor extremely long identity blocks (ICL-500) is optimal for investigating neural face identity representations. While ICL-1 minimizes temporal dependencies, it may hinder the formation of stable identity-related neural representations; conversely, ICL-500 promotes representational stability but

increases susceptibility to long-range contextual effects and block-related confounds. Motivated by this trade-off, BrainFace adopts ICL-50 as its default protocol, providing sufficient consecutive exposures to facilitate stable identity representation learning while mitigating excessive contextual bias.

5.4 Cross-Session Generalization

While the previous experiments demonstrate the existence of decodable identity information, an equally important question is whether such identity-related neural representations remain stable across time.

As illustrated in Fig. 2, each participant completed 10 recording sessions, and each session was acquired independently at a different time point, typically across multiple days. Consequently, the sessions capture natural variations in cognitive state, attention, electrode placement, and recording conditions that inevitably arise during real-world EEG acquisition.

To evaluate temporal robustness, we adopt a leave-one-session-out protocol, where the model is trained on nine sessions and evaluated on the remaining unseen session. Importantly, the held-out session is not a partition of a continuous recording but an independently collected recording acquired on a different day. Therefore, the leave-one-session-out setting evaluates genuine temporal generalization and is substantially more challenging than conventional random train-test splits. Under this protocol, successful decoding requires the learned representations to capture temporally stable identity-related neural information rather than session-specific characteristics, providing a rigorous assessment of the temporal persistence of neural identity representations.

Table 4: Cross-Session identity decoding performance across five subjects.

Metric	S1	S2	S3	S4	S5	Avg.
Top-1 (%)	5.9±0.4	4.3±0.3	4.4±0.2	5.1±0.4	5.5±0.3	5.0
Top-5 (%)	26.9±1.2	18.4±2.5	21.8±1.0	21.7±1.3	23.9±2.1	22.5

The results are summarized in Table 4. Compared with the within-subject mixed-session setting, performance decreases substantially across all participants, with average Top-1 accuracy dropping from 37.5% to 5.0% and Top-5 accuracy decreasing from 85.8% to 22.5%. This performance gap highlights the considerable session-to-session variability inherent in EEG recordings, arising from factors such as changes in cognitive state, attention, electrode placement, and recording conditions across different acquisition days.

Despite this challenge, all participants consistently achieve performance well above the 2.5% chance level. Moreover, EEG-FOFNO remains the best-performing method under the LOSO protocol, indicating that the learned representations retain meaningful identity-related information even when evaluated on independently recorded and temporally unseen sessions.

These findings suggest that identity-related neural representations are neither completely session-specific nor fully invariant over time. Instead, they appear to contain a temporally stable component that persists across recording sessions, while remaining

Table 5: Cross-subject identity decoding performance under the leave-one-subject-out protocol. Results are reported as Top-1 (%) and Top-5 (%) accuracies.

Training → Test	Top-1 (%)	Top-5 (%)
S1, S2, S3, S4 → S5	XX.X _{±X.X}	XX.X _{±X.X}
S1, S2, S3, S5 → S4	XX.X _{±X.X}	XX.X _{±X.X}
S1, S2, S4, S5 → S3	XX.X _{±X.X}	XX.X _{±X.X}
S1, S3, S4, S5 → S2	XX.X _{±X.X}	XX.X _{±X.X}
S2, S3, S4, S5 → S1	XX.X _{±X.X}	XX.X _{±X.X}
Average	XX.X _{±X.X}	XX.X _{±X.X}

affected by substantial session-dependent variability. This observation highlights both the feasibility and the inherent difficulty of long-term EEG-based face identity decoding.

5.5 Cross-Subject Generalization

Beyond temporal stability, an even more challenging question is whether identity-related neural representations are shared across different individuals. To investigate this issue, we adopt a leave-one-subject-out protocol, where the model is trained on four participants and directly evaluated on a previously unseen participant without any subject-specific adaptation.

The results are summarized in Table 5. As expected, performance further decreases compared with the cross-session setting, reflecting the substantial inter-subject variability inherent in EEG signals. Differences in cortical anatomy, neural response patterns, physiological characteristics, and recording conditions make cross-subject transfer particularly challenging [14, 41]. Nevertheless, EEG-FOFNO consistently achieves performance above the 2.5% chance level across all leave-one-subject-out settings and maintains a clear advantage over competing approaches. These results indicate that identity-related information is not entirely subject-specific and that a portion of the learned representations can transfer across individuals.

Taken together, the findings suggest that neural representations of face identity contain both subject-specific and subject-invariant components. While substantial variability remains across individuals, the existence of above-chance cross-subject decoding performance provides evidence for a shared representational structure underlying face identity processing. This result supports the hypothesis that face identity may be encoded, at least partially, through common neural mechanisms across different human subjects.

5.6 Cognitive Factors Analysis

Beyond overall decoding accuracy, we further investigated potential cognitive factors that may influence identity-specific neural responses by focusing on: (i) the gender of the target face and (ii) familiarity between the participant and specific identities.

Gender-Based Performance Patterns. A natural hypothesis is that the gender of faces may systematically affect neural discriminability, potentially leading to consistently higher decoding accuracy for either male or female identities. To examine this, we conducted analyses across six representative models (CNN [20], EEGNet [18], EEGChannelNet [30], EEGConformer [34], DAEeGViT

[29], and EEG-FOFNO). Specifically, we evaluate on: (i) the average recognition rates of male versus female identities across all 40 classes, and (ii) the gender composition of the top 20, top 10, and bottom 10 identities with the highest and lowest recognition accuracy for each model.

Contrary to our initial hypothesis, the results do not exhibit any stable or model-consistent gender advantage. The average recognition performance across all models is nearly identical between male and female identities, where the mean across 6 models between male and female are: 20.16% vs. 20.17%, difference < 0.01%. To further investigate gender-related patterns, we analyzed the composition of the top-performing identities for each model, and the results are illustrated in Fig. 9. As seen, within the Top-20 most accurately recognized identities, males occupy 10.16 on average, and female 9.84. Within the Top-10, males account for 5.17, whereas females account for 4.83. In the Bottom-10 least accurately recognized identities, males and females account for 5.15 and 4.85 on average, respectively. Although these statistics may appear to suggest a slight male advantage, a closer inspection reveals that this tendency is not consistent across models. Specifically, the direction of any minor gender bias is model-dependent rather than stimulus inherent, where as shown in Fig. 9, some models exhibit marginally higher recognition for male identities, while others favor female identities.

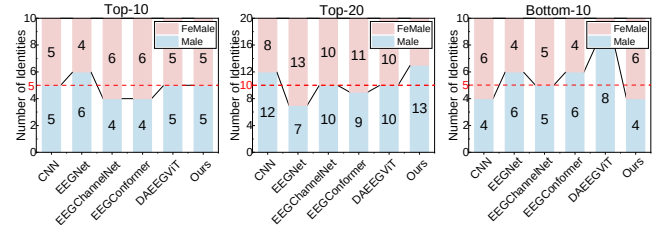


Figure 9: Gender composition of Top-10, Top-20, and Bottom-10 recognized identities across models.

Influence of Face Familiarity. We further examined whether prior familiarity with facial identities is associated with identity-specific EEG decoding performance. After completing all EEG recording sessions, the participant was asked to rate their familiarity with each of the 40 identities on a three-level scale: *unknown*, *familiar*, and *highly familiar*. To avoid biasing the responses, the specific analytical purpose of this familiarity assessment was not disclosed during rating. Among the 40 identities, 3 were reported as unknown, whereas 5 were rated as highly familiar.

By aligning the familiarity ratings with identity-wise decoding performance, we observed a clear correspondence between familiarity and recognition accuracy. Specifically, identities rated as highly familiar consistently appeared in the upper half of the accuracy ranking, whereas identities rated as unknown were located in the lower half. This pattern suggests a positive association between face familiarity and neural discriminability: identities that are more familiar to the participant tend to elicit EEG responses that are more separable for identity decoding. This observation is consistent with cognitive neuroscience findings showing that repeated exposure

and personal familiarity can enhance the stability and separability of face-selective neural representations [2, 40].

These results suggest that familiarity may be an important cognitive factor modulating EEG-based face identity decoding. Rather than being determined solely by the visual appearance of facial stimuli, identity-related neural responses may also be shaped by the observer's prior experience with specific individuals. This finding provides useful guidance for future EEG-based face recognition studies. In particular, when neural signal quality or data consistency is limited, increasing participant familiarity with stimulus identities through controlled pre-exposure or learning-based familiarization may improve representational separability, thereby enhancing decoding reliability and interpretability.

Discussion. Taken together, these results suggest that identity-related neural representations are shaped not only by stimulus-driven visual processing but also by higher-level cognitive factors such as familiarity and prior experience. Although facial identity can be reliably decoded from EEG signals, the underlying representations are not purely determined by visual appearance. Instead, they reflect an interaction between perceptual and cognitive processes. This observation further supports the view that face identity is encoded within a structured neural representational space, whose organization is influenced by both stimulus properties and individual experience. Future work using larger cohorts and controlled familiarity paradigms may help clarify the mechanisms underlying these effects.

5.7 EEG-FOFNO Generalization to Other EEG Benchmarks

The previous experiments demonstrate the effectiveness of EEG-FOFNO for identity-level face recognition and provide evidence that identity-related neural representations exhibit temporal stability and partial cross-subject transferability. An important question, however, is whether the proposed representation learning strategy extends beyond the BrainFace benchmark to more general EEG-based visual decoding tasks.

To investigate this issue, we evaluate EEG-FOFNO on two widely used public benchmarks, including two versions of EEG-ImageNet [1, 35] and ThingsEEG [12].

EEG-ImageNet. Unlike BrainFace, which focuses on fine-grained identity-level face recognition, EEG-ImageNet evaluates coarse-grained category-level visual decoding from EEG responses elicited by diverse object categories, such as animals, tools, and furniture. The results are reported in Table 6. Under the protocol of Spampinato et al. [35], EEG-FOFNO achieves a Top-1 accuracy of 59.5

ThingsEEG. We further evaluate EEG-FOFNO on ThingsEEG [12] under the contrastive decoding framework of NICE-EEG [33], replacing the original EEG encoder with EEG-FOFNO. Similar to EEG-ImageNet, ThingsEEG contains EEG responses elicited by a diverse set of natural object concepts and therefore provides a complementary evaluation of cross-dataset generalization. As shown in Table 9, EEG-FOFNO consistently achieves competitive or superior performance across the majority of subjects and obtains the best average Top-1 and Top-5 accuracies among all compared methods.

Table 6: Comparative listing of the recognition rates achieved on EEG-ImageNet.

Method(%)	Spampinato et al.[35]		Ahmed et al.[1]	
	Top-1	Top-5	Top-1	Top-5
SVM [15]	5.8	18.1	5.0	21.3
KNN [11]	7.4	26.8	2.1	12.7
LSTM [7]	21.4	71.5	2.2	12.4
MLP [37]	6.6	25.1	2.5	12.9
CNN [20]	33.1	78.9	5.1	20.8
EEGNet [18]	31.1	85.3	7.0	24.1
EEGChannelNet [30]	48.1	81.1	2.5	10.3
SyncNet [21]	34.1	86.8	2.5	12.1
EEGConformer [34]	44.1	85.6	10.1	31.9
DAEEGViT [29]	53.1	92.7	11.9	33.1
EEG-FOFNO (Ours)	59.5	96.1	13.3	37.3

Taken together, the results on EEG-ImageNet and ThingsEEG suggest that the representations learned by EEG-FOFNO are not limited to identity-level face decoding. Instead, they transfer effectively across multiple EEG-based visual decoding paradigms, ranging from fine-grained face identity recognition to broader category-level object recognition. These findings indicate that the proposed topology-frequency representation captures neural structures that are useful across diverse visual EEG classification tasks.

5.8 Ablation Study(Please ignore this part of the experiment for now as it is incomplete.)

To evaluate the contribution of each component in EEG-FOFNO, we construct a baseline model by removing the Weighted Fourier Spectral Encoding module from the proposed framework shown in Fig. ??, and conduct ablation experiments on BrainFace-40 under both the mixed-session and leave-one-session-out (LOSO) protocols. The component-level results are summarized in Table 7. With only topology-aware local encoding, the baseline model achieves 36.0% Top-1 accuracy under the mixed-session protocol and 8.8% under LOSO. After introducing the spectral encoding module, the performance increases to 43.1% and 9.3%, respectively, indicating that global frequency-domain modeling provides complementary information beyond local spatiotemporal cues. When combining topology-aware features and spectral representations, replacing the proposed adaptive fusion module with simple concatenation or element-wise addition leads to suboptimal results, yielding 43.1%/9.3% and 43.8%/9.3% under the mixed-session/LOSO protocols, respectively. In contrast, the proposed adaptive fusion module achieves the best performance of 45.8% and 9.8%, demonstrating its effectiveness in dynamically coordinating local spatiotemporal dynamics and global spectral structures. Notably, the same trend is consistently observed under both evaluation protocols, suggesting that the contribution of each component is not limited to the easier mixed-session setting.

We further analyze the effect of the spectral encoding depth in Table 8. Specifically, we vary the number of spectral encoding layers from $L = 1$ to $L = 5$ while keeping the other components unchanged. The best performance is achieved at $L = 2$, reaching 45.8% under the

Table 7: Component ablation of EEG-FOFNO under the mixed-session (Mixed) and leave-one-session-out protocols.

Method	Spectral Encoding	Fusion	Within-Subject	Cross-Subject
Baseline	×	–	36.0	8.8
Variant 1	✓	Concatenation	38.1	9.3
Variant 2	✓	Addition	38.8	9.3
EEG-FOFNO	✓	✓	40.0	9.8

Table 8: Effect of the number of Spectral Encoding layers under the mixed-session (Mixed) and leave-one-session-out (LOSO) protocols.

Method	Number of Layers	Within-Subject	Cross-Subject
EEG-FOFNO ($L = 1$)	1	39.1	9.5
EEG-FOFNO ($L = 2$)	2	40.0	9.8
EEG-FOFNO ($L = 3$)	3	39.3	9.4
EEG-FOFNO ($L = 4$)	4	38.7	9.2
EEG-FOFNO ($L = 5$)	5	37.9	9.2

mixed-session protocol and 9.8% under LOSO. Although shallow spectral modeling already provides clear gains, further increasing the depth does not lead to additional improvements and instead causes a gradual drop in performance.

6 Conclusion

In this work, we introduced BrainFace-40, the first EEG benchmark specifically designed for identity-level face decoding, and conducted a systematic investigation of how facial identity is represented in non-invasive EEG signals. Unlike existing EEG benchmarks that primarily focus on object categories or semantic concepts, BrainFace-40 enables the study of fine-grained identity-related neural representations elicited by face stimuli.

Through extensive experiments, we demonstrate that facial identity can be reliably decoded from EEG signals across multiple participants. Beyond decoding performance, our analyses reveal several important properties of identity-related neural representations. Specifically, identity information remains partially stable across independently recorded sessions, exhibits a transferable component across different individuals, and is influenced by temporal context as well as cognitive factors such as familiarity. These findings suggest that facial identity is encoded not merely through transient neural responses but through a structured neural representation that persists across time and partially generalizes across subjects.

To facilitate the study of this problem, we further propose EEG-FOFNO, a topology-frequency representation learning framework that jointly models topology-aware spatiotemporal dependencies and global spectral dynamics in EEG signals. Extensive evaluations on BrainFace-40, EEG-ImageNet, and ThingsEEG demonstrate that EEG-FOFNO consistently achieves strong performance across diverse EEG-based visual decoding tasks.

Overall, BrainFace-40 and EEG-FOFNO establish a new benchmark and baseline for studying identity-level neural decoding. More broadly, this work provides initial evidence that face identity is represented in a structured and partially transferable neural space, opening new opportunities for understanding neural identity representations and developing more robust EEG-based recognition

systems. Future work will focus on larger multi-subject datasets, more diverse recording conditions, and deeper investigations into the neural mechanisms underlying face identity representation.

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Table 9: Comparative listing of Top-1(Top-5) rates on ThingsEEG across 10 subjects.

Method(%)	Subj 1	Subj 2	Subj 3	Subj 4	Subj 5	Subj 6	Subj 7	Subj 8	Subj 9	Subj 10	AVG
ShallowNet[13]	6.0 (16.0)	3.5 (21.5)	8.5 (28.0)	16.0 (41.0)	4.0 (16.5)	9.5 (32.0)	13.0 (31.0)	10.0 (24.5)	2.5 (9.0)	6.0 (22.5)	7.9 (24.2)
DeepNet[39]	12.0 (31.0)	7.5 (27.5)	10.5 (33.0)	14.0 (36.5)	4.5 (20.0)	10.0 (33.0)	9.5 (30.0)	12.0 (41.0)	9.0 (31.5)	10.0 (33.5)	9.9 (31.7)
EEGConformer[34]	11.0 (39.0)	8.5 (28.0)	11.0 (33.5)	14.5 (38.0)	6.0 (25.5)	10.0 (30.0)	13.0 (40.0)	13.0 (35.5)	10.5 (31.0)	13.5 (37.5)	11.1 (33.8)
EEGNet[18]	11.5 (38.0)	11.0 (35.5)	14.5 (37.5)	15.0 (43.5)	11.0 (30.0)	13.0 (42.5)	14.0 (35.5)	15.0 (42.5)	11.0 (37.0)	14.5 (41.5)	13.1 (38.4)
NICE-EEG[33]	12.3 (36.6)	10.4 (33.9)	13.1 (39.0)	16.4 (47.0)	8.0 (26.9)	14.1 (40.6)	15.2 (42.1)	20.0 (49.9)	13.3 (37.1)	14.9 (41.9)	13.8 (39.5)
ATM[19]	12.1 (36.2)	10.6 (33.7)	13.3 (39.6)	15.9 (43.3)	8.5 (28.1)	14.3 (38.8)	14.7 (40.5)	19.1 (46.3)	13.5 (37.7)	14.5 (41.6)	13.6 (38.6)
EEG-FOFNO(Ours)	13.1 (39.3)	11.6 (36.9)	13.8 (40.2)	16.3 (45.1)	11.1 (32.5)	15.2 (43.8)	14.6 (40.9)	18.3 (48.8)	14.8 (39.5)	15.4 (43.7)	14.4 (41.7)

Table 10: Top-1 and Top-5 accuracy (%) comparison with different methods.

Method	Top-1 (%)	Top-5 (%)
SVM [15]	4.5±0.4	20.7±0.9
KNN [11]	3.2±0.5	16.4±1.1
LSTM [7]	3.5±0.3	19.8±0.6
MLP [37]	3.1±0.4	14.1±0.7
CNN [20]	3.9±0.3	18.9±0.5
EEGChannelNet [30]	4.8±0.3	21.1±0.5
EEGNet [18]	6.1±0.4	24.3±0.7
EEGConformer [34]	7.9±0.4	27.1±0.8
DAEEGViT [29]	7.5±0.6	25.5±0.9
EEG-FOFNO (Ours)	9.8±0.4	29.7±0.7

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